

CPU Test

Rs 12,501 and above

Brand	Intel	AMD	AMD	Intel	AMD	Intel	Intel	Intel
Model	Core 2 Duo E8600	Phenom 2 945	Phenom 2 955BE	Core i7 920	Phenom 2 965BE	Core 2 Quad Q9650	Core 2 Extreme QX9650	Core i7 Extreme 975
Price (Rs) *	13800	13920	14499	14600	14999	16500	52000	56000
Performance (Out of 100)	52.34	58.00	60.72	71.90	63.57	61.67	61.00	90.01
Features								
Socket Interface	LGA 775	AM3/AM2+	AM3/AM2+	LGA 1366	AM3/AM2+	LGA 775	LGA 775	LGA 1366
CPU Codename	Wolfdale	Deneb	Deneb	Nehalem	Deneb	Yorkfield	Yorkfield	Nehalem
Core Frequency	3333 MHz	3000 MHz	3200 MHz	2667 MHz	3400 MHz	3000 MHz	3000 MHz	3333 MHz
Number Of Cores / Hyperthreaded (V/N)	2 / N	4 / N	4 / N	4 / Y	4 / N	4 / N	4 / N	4 / Y
External Bus Frequency	1333 MHz	2000 MHz	2000 MHz	4800	2000 MHz	1333 MHz	1333 MHz	6400
Fabrication Process (in Nanometers)	45 nm	45nm	45nm	45 nm	45nm	45 nm	45 nm	45 nm
L1 Cache / L2 Cache	128 KB / 6 MB	512 KB / 2 MB	512 KB / 2 MB	256 KB / 2 MB	512 KB / 2 MB	256 KB / 12 MB	256 KB / 12 MB	256 KB / 2 MB
L3 Cache (if any)	-	6 MB	6 MB	8 MB	6 MB	-	-	8 MB
Performance								
Synthetic benchmarks								
3D Mark Vantage (CPU Score)	36497	39520	40515	42445	41150	41319	41564	49816
Wprime 2.00 (32 million, in sec) #	22.907	12.167	11.544	9.435	10.81	12.747	12.794	7.273
ScienceMark 2.0 Cryptography (AES, Rijndael) in sec #	9.119	8.56	8.045	10.815	7.601	10.145	10.224	8.513
ScienceMark 2.0 Membench (Memory bandwidth) MBps	6779.02	9507.85	9622.24	11403.95	9717.15	6635.05	6656.65	14054.98
Maxxon CineBench R10 rendering (all cores / one core)	7107 / 3672	9847 / 2748	10492 / 2924	12527 / 3081	11138 / 3106	11848 / 3321	11797 / 3322	16027 / 3871
PovRay 3.62 (Chess2 scene) (in sec) #	161	166	155	171	145	183	183	133
Valve Multicore CPU Particle benchmark	67	87	92	135	97	108	107	170
Real World Applications								
Maya 2008 Rendering all cores / one core (in sec) #	76 / 142	52 / 170	49 / 160	36 / 162	46 / 150	48 / 158	49 / 158	29 / 128
WinRAR 3.9 (all cores / single core)	1802 / 974	2280 / 790	2357 / 812	3980 / 1006	2420 / 833	2548 / 938	2505 / 916	4805 / 1201
DivX 7.0 Encoding (sec) #	170.1	151.7	144.2	113.8	138.5	143.7	144	90
HandBrake x264 encoding (sec) #	319.5	164.6	154.4	118.5	146.3	159.7	161	104.1
Doom 3 (FPS)	296	251.5	265.7	289.8	271.4	290	287.4	361.6
Crysis (CPU benchmark, FPS)	113	93	97	110	98	106	101	130
COH: Tales of Valor (FPS)	568.7	439.3	435.9	502.8	481.2	525.9	518.4	760.2

* Exclusive of all taxes # for these scores lower is better

reckon this figure will scale upwards. Do note, however, that the Core i7 is the only CPU with support for tri-channel memory giving it a strong advantage when it comes to bandwidth-intensive computation. As for rendering, Nehalem's four cores are hyper-threaded and Maya 2008 makes use of all eight (four physical, eight logical) cores rather efficiently and the i7 975 renders our test scene in a mere 29 seconds – the fastest we've ever seen. In fact, any application that supports four cores runs much faster on the Core i7s than on any other CPU. In the DivX encoding test,

the Core i7 975 once again registers the quickest score and is the only CPU to encode a 378-MB VOB file in under 100 seconds.

The high-end tests were an eye opener. Gone is AMD's image of a picker of scraps, it is still the underdog, but has come out fighting with some really competitive options. Gone is the Core 2's dominance as AMD's Phenom IIs finally surpass Intel's Core 2-based dual and quad cores. However, AMD has an uphill fight when it comes to matching toes with the new 1,366 pin-based heavyweight. So the Core i7 975 Extreme is the fastest desktop CPU in the world.

Impressive, but it costs Rs. 56,000 – a small fortune to blow on a processor alone. Surprisingly, it's the cheaper sibling, the i7 920 that steals the limelight – it's cheaper than AMD's fastest CPU and outperforms it by a big margin, and on account of this is our Digit Best Buy. The icing on the cake is that the Core i7 920 with the new D0 stepping is a highly overclockable chip; if the rumours are to be believed it can handle 3.8 GHz on air cooling – impressive indeed. Of course if you want to set records, the i7 975 Extreme with its unlocked multiplier and higher QPI rating – 6.4 giga-transfers versus 4.8

giga-transfers for all other Core i7 chips. Of course you could wait a bit until early next year for the Core i9 processors – these will be based on a 32-nm process and may scale very close to 4 GHz. But then this is all totally pointless to the common man. You have plenty of processing grunt with the current batch of CPUs.

Earlier, AMD had a slight price advantage when it came to its platforms which were DDR2 based. Now with the 790FX and 785G chipsets supporting DDR3 it has lost that. If you desire a powerful system today that can handle the best and